10.2 Finding Arc Measures

Name _____ Period ____ Date _____

Geometry 10.2 (30-minute plan) · Student sheet

I can…

- find the measures of minor arcs, major arcs, and semicircles.
- use the Arc Addition Postulate to find arc measures.
- tell whether two arcs are congruent.

Vocabulary · fill in the missing word

Central angle — an angle whose _____ is the center of the circle

Minor arc — an arc whose _____ is less than 180°

Major arc — an arc whose _____ is greater than 180°

Semicircle — an arc whose _____ are the endpoints of a diameter

Congruent arcs — arcs that have the same _____ and are in the same circle or congruent circles

Part 1 · Notes and examples (fill in each note, then work the example)

Arc measures: a minor arc has the same measure as its _____ angle. A major arc measures _____ minus its minor arc. A semicircle measures 180°.

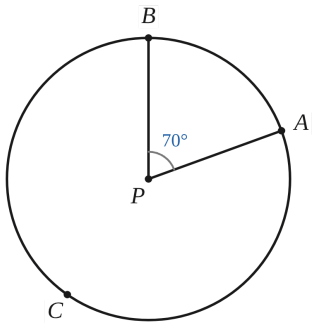
Naming arcs: a minor arc uses its two endpoints; a major arc or a semicircle uses _____ letters.

Example 1

Find the measure of each arc of circle P.

a. arc AB

b. arc ACB



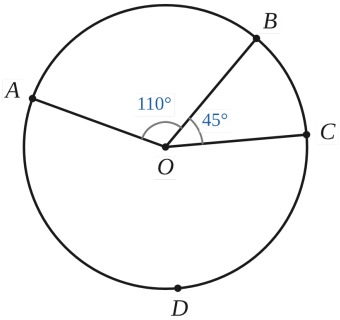
Arc Addition Postulate: the measure of an arc formed by two adjacent arcs is the _____ of the measures of the two arcs.

Example 2

Find the measure of each arc of circle O.

a. arc ABC

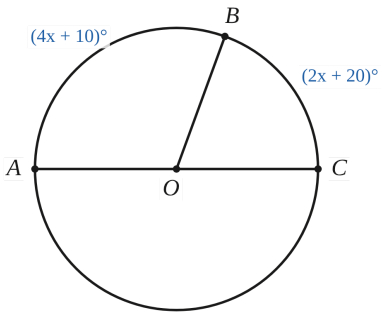
b. arc ADC



Congruent Central Angles Theorem: in the same circle or congruent circles, two minor arcs are congruent if and only if their central angles are _____.

Example 3

$\overline{AC}$ is a diameter of circle O. Find the value of x. Then find m(arc AB).



Part 2 · Partner practice

Partner practice

Three arcs make up a whole circle. They measure (3x)°, (5x + 8)°, and (2x + 12)°. Find x and the measure of each arc.